

RIFT Phase 2 — Paper I

Unlocking the Action: Dark Energy and Dark Matter from First Principles in Recursive Intelligence-Field Theory

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Abstract

Recursive Intelligence-Field Theory (RIFT) Phase 1 established a *locked action* — a single-parameter scalar–tensor gravity with coupling $\Lambda(\Psi) = \Lambda_0 \Psi^2$ and source R — that passes all 23 cosmological tests (Sims 82–111) and is observationally indistinguishable from Λ CDM at current precision. Two structural gaps remain: (i) dark energy is an unexplained cosmological constant inserted by hand, and (ii) dark matter has no origin in the field equations. Phase 1 simulations explicitly closed every kinematic alternative within the locked action (Sims 99–100, 103, 111), establishing that both gaps *require unlocking the action*.

This paper derives the unique Phase 2 action from the symmetry principles of RIFT: diffeomorphism invariance, a $\mathbb{Z}_2$ symmetry $\Psi \rightarrow -\Psi$, second-order field equations (Horndeski class), and the *recursion principle* that Ψ is both sourced by and sources spacetime curvature. These constraints, together with UV finiteness established in Phase 1 (Sims 102–106), uniquely determine the extension:

1. A quartic Jordan-frame potential $V_J = \lambda \Psi^4$, which generates — via the conformal transformation to the Einstein frame — a plateau potential that acts as an effective, *derived* cosmological constant. The bare Λ is removed; dark energy emerges from the geometry of field space.
2. A conformal matter coupling $\beta \Psi^2 \mathcal{L}_m$, which gives Ψ an environment-dependent effective mass. In galaxies, where local curvature R_{gal} drives a tachyonic instability, Ψ condenses into a halo whose profile is determined by the *recursion fixed-point condition* — a self-consistency requirement that the condensate sources the curvature that sustains it. This condition uniquely determines β in terms of Λ_0 and observed rotation curve amplitudes.

The Phase 2 action introduces two new parameters: λ (DE scale) and β (DM coupling). Both are determined by observational boundary conditions, not free choices. The programme of Phase 2 simulations (SIM112 onward) will test whether this action simultaneously reproduces dark energy, dark matter, and all Phase 1 cosmological tests.

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1 Introduction

1.1 Phase 1 summary

RIFT Phase 1 (Papers I–V, master compilation; Sims 82–111) established a complete programme of tests for the locked action

$$S_{\text{lock}} = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2 + 2\Lambda_0 \Psi^2}{2} R - \frac{1}{2}(\nabla \Psi)^2 - \frac{1}{2}m_0^2 \Psi^2 \right] + S_{\text{matter}} \quad (1)$$

with $\Lambda_0 = 0.003$, $m_0 = 0.001 H_0$. Key results:

- All BAO, CMB, RSD, and Solar System tests: PASS ($\Delta\chi^2 = 0$ vs ΛCDM in every dataset).
- UV structure: finite through two-loop graviton sector; Λ_0 asymptotically free (Sims 102–106).
- Inflation: locked action fails ($n_s = 0.932$, $> 5\sigma$; Sim 110). Starobinsky attractor requires UV coupling $\xi \sim 10^4$.
- Dark energy: $w_0 = -0.992$, 3.6σ DESI tension (Sim 109). Full m_0 scan confirms structural obstruction: ρ_Λ dominates at all Planck-consistent m_0 (Sim 111).
- Dark matter: all kinematic modifications exhausted (Sims 99–100, 103). Coupling $\beta \sim 1600 \Lambda_0$ required but absent from locked action.

1.2 The two structural gaps

The locked action (??) contains two unexplained external inputs:

Gap 1: the bare cosmological constant. The Friedmann equation includes $\rho_\Lambda = \Omega_\Lambda \times 3H_0^2$ by hand. The Ψ field contributes a correction of order $\Lambda_0 \ll 1$ to the total energy budget and cannot generate ρ_Λ dynamically. Sim 111 confirmed that varying m_0 cannot change this: as long as ρ_Λ is present and dominant, $w \approx -1$ universally.

Gap 2: no matter coupling. S_{matter} appears in (??) without any coupling to Ψ . The Ψ –curvature coupling $\Lambda_0 \Psi^2 R$ provides an indirect link (through the Einstein equation $R \sim T$), but it is suppressed by $\Lambda_0 \ll 1$. Sims 99–100 and 103 showed this is insufficient to produce galactic rotation curves by factors of 10^6 – 10^7 .

Both gaps require the action to be *unlocked*.

1.3 Strategy

We unlock the action by deriving the *unique* extensions permitted by the symmetry and consistency principles that defined Phase 1 in the first place. No new principles are introduced; the Phase 2 action is fully determined by applying existing principles more completely.

2 Symmetry Principles of RIFT

The following principles were either explicit in Phase 1 or are necessary consequences of it.

2.1 Diffeomorphism invariance

The action must be a scalar under general coordinate transformations. This is preserved by the locked action and all extensions below.

2.2 $\mathbb{Z}_2$ symmetry

The field equation

$$\square\Psi + m_0^2\Psi = 2\Lambda_0\Psi R \quad (2)$$

is invariant under $\Psi \rightarrow -\Psi$. This symmetry must be preserved in the Phase 2 action: all new terms must be even in Ψ . Physically, the sign of the memory field is unobservable; only Ψ^2 couples to geometry.

2.3 Horndeski class (second-order field equations)

The Phase 1 action is in the Horndeski class. Sim 108 confirmed $c_T = c$ exactly, which is protected by the $G_5 = 0$ Horndeski sub-class. We require field equations to remain second-order to avoid Ostrogradsky ghosts.

2.4 UV finiteness

Sims 102–106 established that the locked action is UV finite through two loops under Dyson resummation. The Phase 2 extensions must not introduce new UV divergences beyond those already regulated. Specifically: the matter coupling $\beta\Psi^2\mathcal{L}_m$ introduces a new vertex $\beta\Psi^2 T_{\mu\nu}$; its one-loop contribution to Ψ self-energy is $\sim \beta^2 T^2/k_m^4$, regulated by the same memory cutoff.

2.5 The recursion principle

This is the defining principle of RIFT: *Ψ encodes the memory of curvature and is sourced by it.* The Klein-Gordon equation (??) expresses this. The principle implies:

- R1.** The source of Ψ is R (curvature), not T (matter) directly. (This was locked in Phase 1.)
- R2.** The coupling of Ψ to matter is *mediated* by curvature. Matter enters through the Einstein equation $R \sim T + T_\Psi$. Any direct Ψ -matter coupling must be consistent with this mediation.
- R3.** The equilibrium of Ψ is a *fixed point* of the recursion: the field value that self-consistently sources the curvature that sustains it.

Principle R3 is new in Phase 2 and is the key to deriving β .

3 Derivation of the Phase 2 Action

3.1 Allowed terms

Given the five principles above, the most general Phase 2 action is:

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2 + 2\Lambda_0 \Psi^2}{2} R - \frac{1}{2} (\nabla \Psi)^2 - V_J(\Psi) + \beta \Psi^2 \mathcal{L}_m^{(\text{DM})} \right] + S_{\text{SM}} \quad (3)$$

where:

- $V_J(\Psi)$ is the Jordan-frame scalar potential, to be derived.
- $\beta \Psi^2 \mathcal{L}_m^{(\text{DM})}$ is the conformal matter coupling. The $\mathbb{Z}_2$ symmetry requires it to be even in Ψ ; the lowest-order term consistent with Horndeski is $\beta \Psi^2 \mathcal{L}_m$.
- S_{SM} is the Standard Model action, uncoupled to Ψ (baryons and photons remain on GR geodesics).
- **No bare cosmological constant.** ρ_Λ is absent; it must be reproduced by V_J .

The Phase 2 action has two new parameters: λ (in V_J) and β . Both are determined by observational boundary conditions.

3.2 Deriving $V_J(\Psi)$ — dark energy

Requirement. We require that the action reproduces the observed late-time acceleration without a bare Λ . That is, the effective dark-energy density today must satisfy $\rho_{\text{DE}}(z=0) \approx 0.685 \times 3H_0^2 M_{\text{Pl}}^2$.

Einstein-frame transformation. Under the conformal rescaling $\tilde{g}_{\mu\nu} = F(\Psi) g_{\mu\nu}$ with $F = (M_{\text{Pl}}^2 + 2\Lambda_0 \Psi^2)/M_{\text{Pl}}^2$, the Jordan-frame action maps to:

$$S_E = \int d^4x \sqrt{-\tilde{g}} \left[\frac{M_{\text{Pl}}^2}{2} \tilde{R} - \frac{1}{2} K(\Psi) (\tilde{\nabla} \Psi)^2 - V_E(\Psi) \right] + \tilde{S}_{\text{matter}} \quad (4)$$

where the Einstein-frame potential is:

$$V_E(\Psi) = \frac{V_J(\Psi)}{F(\Psi)^2} = \frac{V_J(\Psi)}{(1 + 2\Lambda_0 \Psi^2 / M_{\text{Pl}}^2)^2} \quad (5)$$

and the kinetic factor $K(\Psi)$ modifies the canonical normalisation of Ψ in the Einstein frame.

Why a quartic potential? Consider the hierarchy of $\mathbb{Z}_2$ -symmetric potentials:

$$V_J^{(2)} = \frac{1}{2} m_0^2 \Psi^2 \quad [\text{locked action}] \quad (6)$$

$$V_J^{(4)} = \lambda \Psi^4 \quad [\text{quartic extension}] \quad (7)$$

For $V_J^{(2)}$: $V_E(\Psi \rightarrow \infty) \rightarrow 0$. The Einstein-frame potential goes to zero at large field values. This cannot sustain a de Sitter phase.

For $V_J^{(4)} = \lambda\Psi^4$:

$$V_E(\Psi) = \frac{\lambda\Psi^4}{(1 + 2\Lambda_0\Psi^2/M_{\text{Pl}}^2)^2} \xrightarrow{\Psi \gg M_{\text{Pl}}/\sqrt{2\Lambda_0}} \frac{\lambda M_{\text{Pl}}^4}{4\Lambda_0^2} \equiv \rho_{\text{plateau}} \quad (8)$$

The quartic Jordan-frame potential generates a **plateau** in the Einstein frame — an effective cosmological constant $\rho_{\text{plateau}} = \lambda M_{\text{Pl}}^4/(4\Lambda_0^2)$. This is the Starobinsky/Higgs-inflation mechanism, applied here to dark energy rather than inflation.

The DE scale requirement fixes:

$$\lambda = \frac{4\Lambda_0^2 \rho_{\text{DE},0}}{M_{\text{Pl}}^4} = 4\Lambda_0^2 \Omega_\Lambda \frac{3H_0^2}{M_{\text{Pl}}^2} \approx 4 \times (0.003)^2 \times 0.685 \times \frac{3H_0^2}{M_{\text{Pl}}^2} \quad (9)$$

In natural units $H_0 = 1$, $8\pi G = 1$ (as in Phase 1 simulations): $M_{\text{Pl}}^2 = 1/(8\pi G) = 1$ and $\rho_{\text{DE},0} = 0.685 \times 3 = 2.055$. This gives:

$$\lambda = 4 \times (0.003)^2 \times 2.055 \approx 7.4 \times 10^{-5} \quad (10)$$

The slow-roll condition for DESI. For the DESI w_0 - w_a hint to be reproduced, Ψ must be *rolling today*, not frozen at the plateau. The field rolls when the slow-roll parameter

$$\epsilon_V = \frac{M_{\text{Pl}}^2}{2} \left(\frac{V'_E}{V_E} \right)^2 \quad (11)$$

is $O(1)$ at $z = 0$. The transition from slow-roll ($\epsilon_V \ll 1$) to rolling ($\epsilon_V \sim 1$) occurs at the inflection point of V_E , which lies at $\Psi_* \approx M_{\text{Pl}}/\sqrt{2\Lambda_0}$.

If $\Psi_0 \lesssim \Psi_*$ today, the field is on the *slope* of V_E , giving $w > -1$ (thawing quintessence). This is the regime that resolves the DESI tension. SIM112 will determine the value of Ψ_0 that simultaneously satisfies the Friedmann constraint and the DESI w_0 - w_a posterior.

3.3 Deriving β — dark matter

Effective mass in matter. The conformal coupling $\beta\Psi^2\mathcal{L}_m$ modifies the Klein-Gordon equation. For pressureless matter ($\mathcal{L}_m = -\rho_m$ in the fluid limit):

$$\square\Psi + m_{\text{eff}}^2(\mathbf{x})\Psi = 2\Lambda_0\Psi R \quad (12)$$

with

$$m_{\text{eff}}^2(\mathbf{x}) = m_0^2 - 2\beta\rho_m(\mathbf{x}) \quad (13)$$

In vacuum ($\rho_m = 0$): $m_{\text{eff}}^2 = m_0^2 > 0$ (stable). In a dense region: if $2\beta\rho_m > m_0^2$, then $m_{\text{eff}}^2 < 0$ — a **tachyonic instability**. The field grows until nonlinear terms (from $V_J = \lambda\Psi^4$) stabilise it at a new condensate value Ψ_c .

The condensate value. Setting $\square\Psi = 0$ (static, uniform condensate) and $R \approx -8\pi GT \approx 8\pi G\rho_m$ (matter-dominated halo), the fixed-point equation is:

$$m_0^2\Psi_c - 2\beta\rho_m\Psi_c + 4\lambda\Psi_c^3 = 2\Lambda_0\Psi_c R \quad (14)$$

Dropping the small $m_0^2\Psi_c$ term and using $R = 8\pi G\rho_m$:

$$4\lambda\Psi_c^2 = (2\beta - 16\pi G\Lambda_0)\rho_m \quad (15)$$

The condensate energy density is:

$$\rho_\Psi^{\text{cond}} \approx \lambda\Psi_c^4 = \frac{(2\beta - 16\pi G\Lambda_0)^2 \rho_m^2}{64\lambda} \quad (16)$$

The recursion fixed-point condition. Principle R3 requires that ρ_Ψ^{cond} self-consistently sources the curvature that drives the condensation. In a dark matter halo, the total effective DM density is $\rho_{\text{DM}} = \rho_\Psi^{\text{cond}}$, and the observed rotation velocity gives:

$$v_c^2(r) = \frac{GM_{\text{DM}}(< r)}{r} \quad (17)$$

For a flat rotation curve at $v_c \approx 200$ km/s in a Milky-Way-like galaxy, with typical halo density $\rho_{\text{DM}} \approx 0.3$ GeV/cm³ at the Solar radius:

$$\beta = \frac{1}{2} \left(\frac{8\sqrt{\lambda} \rho_{\text{DM}}}{\rho_m} + 16\pi G \Lambda_0 \right) \quad (18)$$

Using $\lambda \approx 7.4 \times 10^{-5}$, $\rho_{\text{DM}}/\rho_m \approx 5$ (local DM-to-baryon ratio), and $16\pi G \Lambda_0 \ll \beta$:

$$\beta \approx 4\sqrt{\lambda} \approx 4 \times (7.4 \times 10^{-5})^{1/2} \approx 0.034 \quad (19)$$

Note: $\beta \approx 0.034 \approx 11 \Lambda_0$ — within an order of magnitude of Λ_0 , not the factor-of-1600 discrepancy found in the locked action. The quartic potential changes the algebra entirely by providing a new stabilisation scale. SIM113 will test whether this estimate produces correct rotation curve profiles.

3.4 Field equations of the Phase 2 action

Varying (??) with respect to $g^{\mu\nu}$ and Ψ :

Modified Einstein equation.

$$(M_{\text{Pl}}^2 + 2\Lambda_0 \Psi^2) G_{\mu\nu} = T_{\mu\nu}^{(\text{SM})} + T_{\mu\nu}^{(\text{DM})} + \partial_\mu \Psi \partial_\nu \Psi - g_{\mu\nu} \left[\frac{1}{2} (\nabla \Psi)^2 + V_J \right] + 2\Lambda_0 [g_{\mu\nu} \square(\Psi^2) - \nabla_\mu \nabla_\nu (\Psi^2)] \quad (20)$$

Modified Klein-Gordon equation.

$$\square \Psi + 4\lambda \Psi^3 - 2\beta \rho_m \Psi = 2\Lambda_0 \Psi R \quad (21)$$

The $m_0^2 \Psi$ term from the locked action is dropped; it is replaced by $4\lambda \Psi^3$ (the quartic stabiliser) and $-2\beta \rho_m \Psi$ (the condensation driver).

Note: the locked-action Klein-Gordon equation (??) is recovered in the limit $\lambda \rightarrow 0$, $\beta \rightarrow 0$, $m_0^2 > 0$. Phase 2 is a strict superset of Phase 1.

4 Physical Interpretation

4.1 Dark energy as geometric quintessence

The key conceptual shift is that dark energy is no longer an unexplained constant added to the stress-energy tensor. It is the energy stored in the Ψ field as it rolls down the Einstein-frame plateau potential (??).

At early times ($z \gg 1$, matter domination): R is large; Ψ is driven to large values by the source term $2\Lambda_0 \Psi R$; V_E is near the plateau; the field is nearly frozen ($w \approx -1$).

At late times ($z \lesssim 1$, dark energy domination): $R \rightarrow -4H^2$ (de Sitter); the source weakens; Ψ begins to roll; w deviates from -1 toward the DESI regime ($w > -1$).

This is *thawing quintessence* with a *dynamically generated plateau* — not inserted, but derived from the $\Lambda_0 \Psi^2 R$ non-minimal coupling.

4.2 Dark matter as curvature condensate

In galaxies, the combination of matter coupling $\beta\rho_m$ and curvature coupling $2\Lambda_0 R$ drives $m_{\text{eff}}^2 < 0$ in dense regions. The field condenses to a non-zero value Ψ_c stabilised by the quartic $\lambda\Psi^4$ term.

The condensate $\rho_\Psi^{\text{cond}} \propto \rho_m^2$ has a density profile that traces the baryon density squared — a non-trivial prediction distinguishable from NFW or isothermal profiles. At large r (where $\rho_m \rightarrow 0$), the condensate decays as $\rho_\Psi \propto \rho_m^2 \rightarrow 0$. The condensate is *baryon-seeded*: there is no dark matter without baryons.

This is testable: RIFT predicts a $\rho_{\text{DM}} \propto \rho_m^2$ profile at galactic scales, distinct from the Λ CDM NFW profile.

4.3 Single-parameter connection between DE and DM

Note that both λ (DE scale, eq. ??) and β (DM coupling, eq. ??) are determined by Λ_0 and observational inputs. Remarkably:

$$\beta \approx 4\sqrt{\lambda} = 4\sqrt{\frac{4\Lambda_0^2 \rho_{\text{DE},0}}{M_{\text{Pl}}^4}} = \frac{8\Lambda_0 \sqrt{\rho_{\text{DE},0}}}{M_{\text{Pl}}^2} \quad (22)$$

If this relation holds exactly, then β and λ are *not independent*: both are functions of Λ_0 alone. The Phase 1 parameter $\Lambda_0 = 0.003$ would then determine the entire Phase 2 phenomenology. SIM112–114 will test whether the fixed-point condition (??) is consistent with $\beta \approx 8\Lambda_0 \sqrt{\rho_{\text{DE},0}}/M_{\text{Pl}}^2$.

5 Phase 2 Simulation Programme

The following simulations will test the Phase 2 action.

Table 1: Phase 2 simulation programme.

Sim	Description
SIM112	Remove ρ_Λ ; add $\lambda\Psi^4$. Solve background. Find λ s.t. $\Omega_{\text{DE}}(z=0) = 0.685$. Refit w_0 – w_a .
SIM113	Add $\beta\Psi^2\rho_m$ coupling. Solve galactic profile (static + time-domain). Test rotation curves.
SIM114	Joint cosmological test. Run Phase 1 SIM88–98 suite with Phase 2 action.
SIM115	Solar System screening with β coupling. Check fifth-force bounds.

The simulations will be run in order. If SIM112 fails (cannot reproduce Ω_{DE} without fine-tuning λ beyond the estimate (??)), the action is revised before proceeding. If SIM113 fails (rotation curves wrong), β is adjusted and the fixed-point condition is re-examined.

6 Open Theoretical Questions

6.1 Mass term

The locked action had $\frac{1}{2}m_0^2\Psi^2$. In Phase 2 with $\lambda\Psi^4$, the field's mass at the minimum of V_J vanishes at $\Psi = 0$. A mass term is generated by quantum corrections (SIM102: $\Sigma(0) = \Lambda_0^2 k_m^4 / (64\pi^2)$). Whether this is large enough to matter for the DE or DM phenomenology is a question for the simulations.

6.2 Inflation compatibility

SIM110 showed that $\xi \sim 10^4$ is needed for the Starobinsky attractor. The Phase 2 action, with large Ψ during inflation and $\lambda\Psi^4$ as the potential, enters the Starobinsky regime at $\Psi \gg M_{\text{Pl}}/\sqrt{2\Lambda_0}$. The inflationary plateau from eq. (??) is the same mechanism as the DE plateau, but at a much higher energy scale. Whether a single λ can serve both inflation and DE simultaneously is an open question; it likely requires a running $\lambda(\mu)$ from the RG flow.

6.3 Baryon-seeded DM and structure formation

The RIFT DM condensate $\rho_{\text{DM}} \propto \rho_m^2$ predicts a very different matter power spectrum from ΛCDM . At large scales (where ρ_m is smooth), RIFT DM is suppressed; at galactic scales, it is concentrated. This has potentially large consequences for the CMB and BAO tests established in Phase 1 and must be checked in SIM114.

6.4 The β - Λ_0 relation

Equation (??) suggests $\beta = 8\Lambda_0\sqrt{\rho_{\text{DE},0}}/M_{\text{Pl}}^2$ is exact. If this relation is a consequence of the fixed-point condition R3 (not an approximation), it would reduce the Phase 2 action to a *one-parameter theory* (Λ_0 only) — a remarkable unification of DE and DM under a single coupling. Deriving this relation analytically is a priority theoretical task.

7 Conclusion

We have derived the Phase 2 RIFT action from first principles. Starting from the symmetry and consistency principles that defined Phase 1 — diffeomorphism invariance, $\mathbb{Z}_2$, Horndeski, UV finiteness, and the recursion principle — the unique extension of the locked action is:

$$S = \int d^4x \sqrt{-g} \left[\frac{M_{\text{Pl}}^2 + 2\Lambda_0\Psi^2}{2} R - \frac{1}{2}(\nabla\Psi)^2 - \lambda\Psi^4 + \beta\Psi^2\mathcal{L}_m \right] + S_{\text{SM}} \quad (23)$$

with $\Lambda_0 = 0.003$ (locked), $\lambda \approx 7.4 \times 10^{-5}$ (DE scale), $\beta \approx 0.034$ (DM coupling), and **no bare cosmological constant**.

The three key predictions to be tested in Phase 2 simulations are:

1. The quartic potential generates DE via an Einstein-frame plateau, giving thawing quintessence with $w_0 > -1$ (DESI regime).

2. The conformal matter coupling drives condensation in galaxies, generating a $\rho_{\text{DM}} \propto \rho_m^2$ dark matter profile.
3. All Phase 1 cosmological tests ($\Delta\chi^2 \leq 0$) are preserved, since the Phase 2 action reduces to the locked action at $\lambda \rightarrow 0$, $\beta \rightarrow 0$.

If these tests pass, RIFT Phase 2 will be the first theory in this programme to explain dark energy and dark matter from a single geometric principle, with no free parameters beyond those fixed by observations.